

MATRIX

Rank of a matrix

Submatrices of a matrix

Let A be a $m \times n$ matrix, then a matrix obtained by deleting some rows and columns of A is called submatrix of A .

Minors of a matrix - Every square matrix possesses a determinant. If A be an $m \times n$ matrix, then the determinant of every square submatrix of A is called a minor of the matrix A .

Rank of matrix - The rank of a matrix is the order of the largest square submatrix, whose determinant is not equal to zero.

An n^2 matrix A , is called non singular, if its rank $= n$, i.e. $|A| \neq 0$. Otherwise A is called singular.

We assign the rank 0 to every null matrix. The rank of every non singular matrix of order n is n .

Thus a number r is said to be the rank of a matrix A if it satisfies the following properties:

(i) There is at least one square sub-matrix of A of order r whose determinant is not equal to zero.

(ii) If the matrix A contains any square sub-matrix of order $r + 1$, then the determinant of every such square sub-matrix of order $r + 1$ should be zero.

Remark. The rank of a null matrix is defined as zero.

Q2.

$$\begin{bmatrix} 2 & 4 \\ 3 & 3 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 2 & 4 \\ 3 & 3 \end{bmatrix}$$

$$|A| = 6 - 12 = -6 \neq 0$$

$$\text{Rank} = \underline{\underline{2.}}$$

Q3

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}$$

$$|A| = 1 \times (6 - 8) - 2(4) + 3(4)$$

$$= -2 - 8 + 12$$

$$= -10 + 12$$

$$= \underline{\underline{2}} \neq 0$$

$$\text{Rank} = 3$$

$$Q4. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

$$\begin{aligned} |A| &= 1(21 - 20) - 2(14 - 12) + 3(10 - 9) \\ &= 1 - 2(2) + 3(1) \\ &= 1 - 4 + 3 \\ &= 0 \end{aligned}$$

$$\text{let } A_1 \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = |A_1| = 3 - 4 = -1 \neq 0.$$

$$\therefore \text{Rank of } A = 2$$

$$Q5. \begin{bmatrix} -2 & 1 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 7 \end{bmatrix}$$

$$|A| = -28$$

$$\text{let } A_1 = \begin{bmatrix} -2 & 1 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\begin{aligned} |A_1| &= -2(4 - 3) - 1(0 - 1) + 3(-1) \\ &= -2(1) - 1(-1) + -3 \\ &= -2 + 1 - 3 \\ &= -4 \neq 0. \end{aligned}$$

$$\therefore \text{Rank of } A = 3$$

Elementary Transformations

An elementary transformation is an operation of any one of the following three types.

(i) The interchange of any two rows or columns.

$$R_i \leftrightarrow R_j \quad R_{ij}$$

$$C_i \leftrightarrow C_j \quad C_{ij}$$

(ii) The multiplication of the element of any row or column by any non zero no.

$$R_i \rightarrow KR_i \quad R_i(K)$$

$$C_i \rightarrow KC_i \quad C_i(K)$$

(iii) The addition to the elements of any row or column, the corresponding elements of any other row or column by any non-zero number.

$$R_i \rightarrow R_i + KR_j, \quad R_{ij}(K)$$

$$C_i \rightarrow C_i + KC_j, \quad C_{ij}(K)$$

An elementary transformation is called row transformation or column transformation according as it applies to rows and columns.

Equivalent Matrices

If B be a matrix obtained from A by a finite sequence of elementary transformation on A , then A is said to be equivalent to B which is read by

$$A \sim B.$$

Elementary transformations do not change the rank of a matrix. i.e. Equivalent matrices have the same rank.

The Normal Form of Matrix

Every $m \times n$ matrix of rank r can be reduced to the form by a sequence of elementary transformations

$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ where I_r is the unit matrix of order r and each 0 is a zero matrix or null matrix of appropriate order. The above form to which a given matrix of rank r can be reduced is known as the

First form or Normal Form of canonical matrix

A zero matrix is its own normal. To reduce a non zero matrix to its normal form, both elementary row transformation and elementary column transformation can be used.

Q1. Find the rank of the matrix by reducing to its normal form.

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$C_2 \rightarrow C_2 - 2C_1$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$C_2 \leftrightarrow C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$R_2 \rightarrow -R_2$$

$$\sim \begin{bmatrix} I_2 & 0 \end{bmatrix}$$

$$\therefore \text{Rank } A = \underline{\underline{2}}$$

Q2.

$$\text{Let } A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$$C_1 \leftrightarrow C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 0 & 2 & 1 & 3 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 2 & 6 \\ 0 & 2 & 1 & 3 \end{bmatrix}$$

$$\begin{aligned} C_3 &\rightarrow C_3 - 2C_1 \\ C_4 &\rightarrow C_4 - 2C_1 \end{aligned}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 4 & 6 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$C_2 \leftrightarrow C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2} R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} \therefore \text{Rank of } A = 2$$

Let $A = \begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & -1 & 3 & 6 \\ 0 & 4 & -6 & -10 \\ 5 & 3 & 3 & 11 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 3 & 6 \\ 0 & 4 & -6 & -10 \\ 0 & 8 & -12 & -19 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & -6 & -10 \\ 0 & 8 & 12 & -19 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & 4 & -10 \\ 0 & -12 & 8 & -19 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & -10 \\ 0 & 2 & 8 & -19 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 4 & -10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - 2C_2$$

$$C_4 \rightarrow C_4 - 3C_2$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 5R_1$$

$$3 + 15$$

$$3 - 15$$

$$11 - 30$$

$$C_2 \rightarrow C_2 + C_1$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$C_4 \rightarrow C_4 - 6C_1$$

$$C_3 \leftrightarrow C_2$$

$$C_2 \rightarrow \frac{1}{6} C_2$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$-19 - 2 \times -10$$

$$-19 + 20$$

$$C_4 \leftrightarrow C_3$$

$$C_3 \rightarrow C_3 + 10C_2$$

$$C_4 \rightarrow C_4 - 4C_2$$

$$\sim \begin{bmatrix} I_3 & 0 \end{bmatrix}$$

$$\therefore \text{Rank of } A = \underline{\underline{3}}$$

H.W
(i)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

(ii)

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$